

MERSING, Malaysia

WMO: 486740

Lat: 2.4450N

Lon: 103.8310E

Elev: 45

StdP: 100.79

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.0	22.8	20.8	15.6	24.6	21.0	15.8	24.5	8.4	26.9	7.8	27.0	2.5	240	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.8	34.2	28.5	33.8	28.4	33.1	28.0	29.3	33.3	28.9	32.9	28.5	32.3	3.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.5	32.8	27.9	24.1	32.6	27.2	23.1	31.7	96.4	33.6	94.6	32.6	92.2	33.0	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.3	5.6	21.4	35.3	0.7	1.3	20.9	36.2	20.4	37.0	20.0	37.7	19.5	38.6		
(5)			20.7	30.4	0.6	0.7	20.3	30.9	19.9	31.3	19.5	31.7	19.1	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.8	27.3	27.7	28.2	28.5	28.2	27.9	27.8	27.7	27.6	27.2	26.7	(6)
	DBStd	1.18	1.30	1.11	1.20	1.07	1.04	1.05	0.98	0.90	0.86	0.92	0.98	1.20	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6437	522	486	549	544	573	547	555	551	531	546	514	519	(10)
	CDD18.3	3396	264	252	291	294	315	297	297	293	281	288	264	261	(11)
	CDH23.3	32291	2412	2513	3041	3089	3201	2896	2745	2658	2498	2594	2325	2319	(12)
(13)	CDH26.7	11444	537	671	1075	1268	1359	1182	1067	1026	974	993	735	557	(13)
(14)	Wind	WSAvg	2.8	3.9	3.9	3.0	2.3	2.3	2.4	2.5	2.6	2.4	2.3	3.3	(14)
(15) Precipitation	PrecAvg	2623	321	128	133	121	145	146	163	173	165	203	350	606	(15)
	PrecMax	3674	880	497	361	331	328	287	309	323	340	429	641	1536	(16)
	PrecMin	1672	4	1	0	5	47	22	0	62	3	52	33	73	(17)
	PrecStd	537	230	130	90	79	64	53	62	55	80	89	151	327	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.1	33.1	35.0	35.8	34.9	34.1	34.2	33.9	34.0	33.0	32.2	(19)
		MCWB	27.3	27.0	27.9	29.6	29.2	27.9	28.5	27.6	28.3	28.2	27.8	27.1	(20)
	2%	DB	31.0	31.2	32.1	33.9	34.2	33.9	33.1	33.1	33.0	33.0	32.0	31.1	(21)
		MCWB	26.9	26.6	27.1	28.7	28.8	28.2	28.0	27.8	28.0	27.8	27.2	26.9	(22)
	5%	DB	30.0	30.8	31.8	32.9	33.2	33.0	32.2	32.2	32.2	32.1	31.1	30.1	(23)
		MCWB	26.1	26.4	26.9	28.0	28.3	28.0	27.6	27.5	27.5	27.4	27.0	26.3	(24)
	10%	DB	29.1	30.0	31.0	32.0	32.2	32.1	31.8	31.2	31.2	31.2	30.2	29.2	(25)
		MCWB	25.5	25.9	26.6	27.5	27.8	27.6	27.5	27.3	27.2	27.2	26.5	25.8	(26)
	0.4%	WB	27.9	27.5	28.5	29.7	29.8	29.7	29.4	29.1	29.1	29.0	28.2	27.9	(27)
		MCDB	31.0	31.2	32.1	34.3	34.2	33.5	33.2	32.6	32.8	32.8	31.8	31.1	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.0	26.8	27.6	29.0	29.2	29.0	28.7	28.4	28.5	28.2	27.6	27.1	(29)
		MCDB	30.2	30.3	31.4	33.2	33.5	32.9	32.4	32.0	32.3	32.1	31.0	30.1	(30)
	5%	WB	26.4	26.3	27.1	28.3	28.7	28.3	28.1	28.0	28.0	27.8	27.1	26.5	(31)
		MCDB	29.0	29.6	30.8	32.2	32.9	32.1	31.7	31.6	31.8	31.6	30.5	29.2	(32)
	10%	WB	26.0	26.0	26.5	27.7	28.1	27.8	27.5	27.4	27.4	27.2	26.7	26.1	(33)
		MCDB	28.3	29.1	30.0	31.4	32.0	31.3	31.0	30.9	31.0	30.8	29.7	28.6	(34)
	5% DB	MDBR	4.4	5.2	6.5	8.1	8.8	8.5	8.4	8.4	8.5	8.2	6.8	5.1	(35)
		MCDBR	5.8	6.2	7.5	9.2	9.7	9.4	9.1	9.1	9.2	9.1	8.0	6.5	(36)
		MCWBR	3.1	3.0	3.7	4.8	5.1	4.9	5.1	5.1	5.2	5.0	4.2	3.4	(37)
		MCWBR	5.5	5.7	7.0	8.9	9.3	9.0	8.9	8.7	9.0	8.8	7.5	6.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.461	0.460	0.464	0.464	0.479	0.501	0.491	0.526	0.547	0.557	0.457	0.463	(40)
		taud	2.321	2.309	2.301	2.299	2.262	2.196	2.236	2.118	2.057	2.013	2.334	2.325	(41)
	Ebn at Noon		867	874	866	848	813	783	797	786	785	781	864	858	(42)
		Edh at Noon	133	137	138	135	136	142	138	160	173	182	130	131	(43)
(44) All-Sky Solar Radiation	RadAvg		4.33	5.18	5.30	5.33	5.01	4.72	4.71	4.82	4.79	4.75	4.25	3.96	(44)
	RadStd		0.39	0.47	0.45	0.27	0.22	0.26	0.29	0.20	0.28	0.33	0.24	0.38	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.33	+0.10	N/A	+0.24	+0.35	+0.32	N/A	N/A	+90	+88

Nomenclature: See separate page